

TED (15) 6212
(REVISION-2015)

Register No:.....

Signature:.....

DIPLOMA EXAMINATION IN ENGINEERING / TECHNOLOGY/
MANAGEMENT/COMMERCIAL PRACTICE
ANALYTICAL INSTRUMENTATION

MODEL QUESTION PAPER

[Time: 3 hours]

(Maximum marks: 100)

PART - A

(Maximum 10 Marks)

Marks

I. Answer **all** questions in one or two sentences. Each question carries 2 marks.

1. State Lamberts law of photometry?
2. Name any four basic components of flame photometer?
3. Define Ph and ph scale ?
4. List the classification of chromatography ?
5. Define thermal conductivity of gases?

(5x2=10)

PART - B

(Maximum 30 Marks)

II. Answer **any five** of the following questions. Each question carries 6 marks.

1. Describe the construction and working of single beam filter photometer?
2. List the basic components of Infrared spectrometer?
3. Describe the construction of Raman spectrometer ?
4. Explain ph control in effluent treatment ?
6. Describe the working of paper chromatography?
7. State the principle of electrical conductivity anslzyer?

(5x6=30)

PART - C

(Maximum 60 Marks)

Marks

(Answer **one full** question from each unit. Each full question carries 15 marks.)

UNIT - I

III. a) Explain the construction and working principle of double beam filter photometer? 8

b) Draw and explain the construction and working of infrared spectrophotometer? 7

OR

IV. Draw and explain the construction and working of single beam and double beam spectrophotometer? 15

UNIT - II

V. a) Draw and explain the construction and working of flame photometer? 15

OR

VI. Describe the principle, construction and working of NMR spectrometer? 15

UNIT -III

VII. a) Describe the importance of buffer solution? 6

b) Draw and explain the construction and working of glass electrode 9

OR

VIII. a) Draw and explain the construction and working of calomel electrode? 7

b) Explain dip and flow type industrial electrode assembly? 8

UNIT - IV

IX. a) State the principle of paramagnetic analyzer 3

b) Explain the construction and operation of magnetic force type and wind type paramagnetic oxygen analyzer? 12

OR

X. a) Classify of infrared analyzers? 3

b) Explain the construction and operation of positive and negative infrared analyzer? 12

